UNIVERSITY OF PENNSYLVANIA ESE 5700

The 6T SRAM bitcell

Project 1, Part 1: design, analysis and present evidence

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Cadence GPDK 18 nm $V_{DD} = 0.8$ V Schematic β sweep recorded; post-layout characterization pending

Executive summary

The symmetric 6T cell was drawn in Virtuoso. Read disturb and write contention were analyzed by hand, and three schematic read butterflies were exported from Spectre. At 0.8 V, increasing β from 1 to 2 raises read SNM from 126.6 to 156.3 mV and lowers the disturbed zero-node crossing from 134.3 to 77.1 mV. The $\beta = 2$, $\gamma = 1$ cell is a schematic candidate. Layout signoff, extraction, and post-layout characterization remain incomplete.

Table 1: What this report can establish now. Values are from `results/part1/read-snm/beta*/snm.json`.

β	PD / PG / PU fins	Schematic read SNM (mV)	Read-disturbed zero node (mV)
1.0	2 / 2 / 2	126.6	134.3
1.5	3 / 2 / 2	145.8	96.6
2.0	**4 / 2 / 2**	**156.3**	**77.1**

1 Assignment scope and evidence standard

The current handout requires: sized symmetric schematic and hand analysis; a three-point schematic β sweep; custom tileable layout with area and DRC/LVS summaries; per-cell BL and WL capacitance after extraction; post-layout hold/read SNM, write margin, cell current and supply dependence; a transient write/read sequence; and a discussion of the limiting trade-off. It also requires individual team contributions. Figures and numbers below are labeled as *measured schematic*, *hand estimate*, or *open*. The supplied handout sets $F = 10$ nm for area in F^2, and supply points of 0.8, 0.7 and 0.5 V.

2 Cell schematic and selected dimensions

A 6T cell consists of cross-coupled inverters INV1 and INV2 and two pass-gates. Both storage nodes connect to precharged bitlines only while WL is high. For the candidate schematic, the captured device settings are $L = 18$ nm and $n_{\text{fin}} = 4$ for each pull-down, $n_{\text{fin}} = 2$ for each pull-up, and $n_{\text{fin}} = 2$ for each pass-gate. Thus the nominal fin-count ratios are

$$\beta = \frac{(W/L)_{\text{PD}}}{(W/L)_{\text{PG}}} \simeq \frac{4}{2} = 2, \qquad \gamma = \frac{(W/L)_{\text{PU}}}{(W/L)_{\text{PG}}} \simeq \frac{2}{2} = 1. \tag{1}$$

The effective process-model strengths are not simply fin counts; the ratios express the intended relative sizing. Both halves are identical, meeting the symmetric-cell requirement at schematic level.

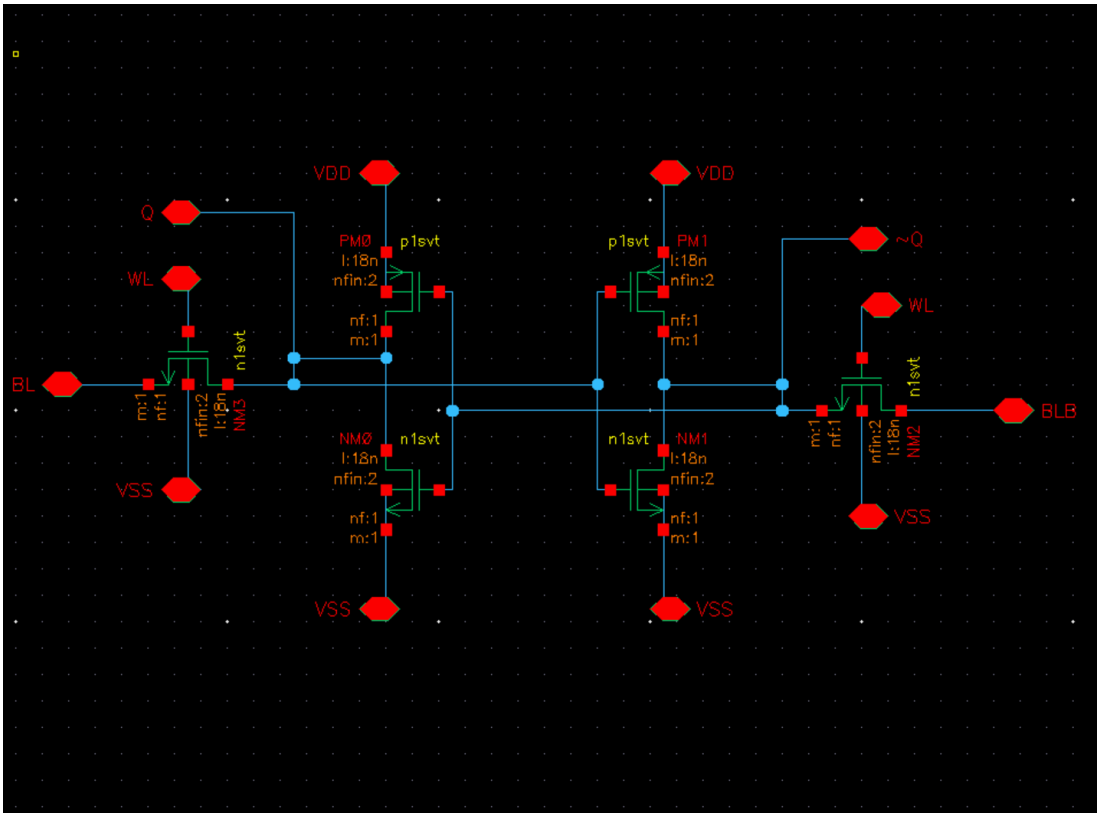


Figure 1: Original Virtuoso cell schematic, with device fin counts and 18 nm length visible. The plot and bench evidence in later sections comes from this family of schematics.

3 Hand sizing analysis

3.1 Read disturb and the cell ratio

Assume $Q = 0$, $\overline{Q} \approx V_{DD}$, WL= V_{DD} and BL=BLB= V_{DD}. Current flows from BL through the pass-gate into Q and down through the pull-down. The handwritten circuit and divider in Figure 2 were part of the original analysis. They identify the two opposing devices and the node whose rise can upset INV2.

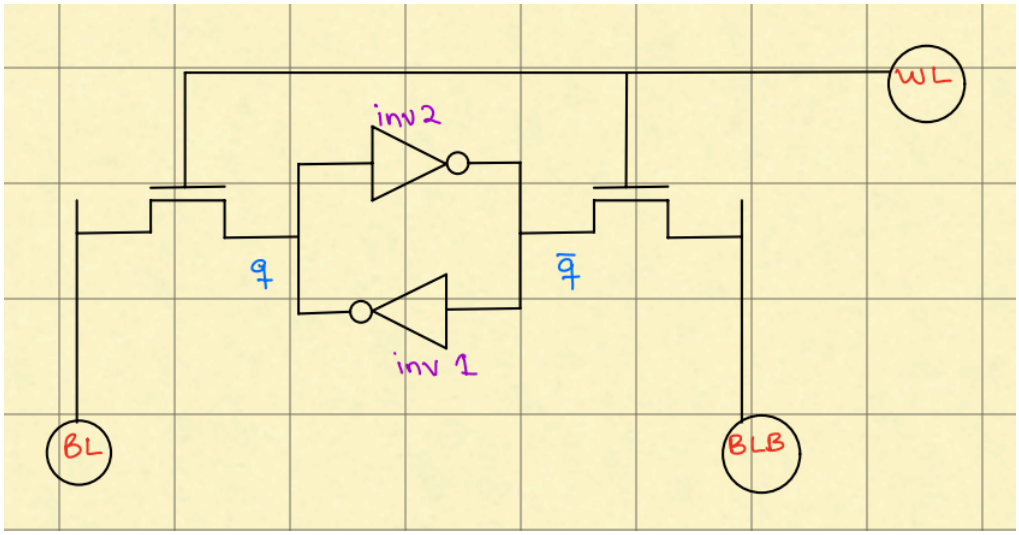


Figure 2: Original handwritten read-state diagram, with node labels.

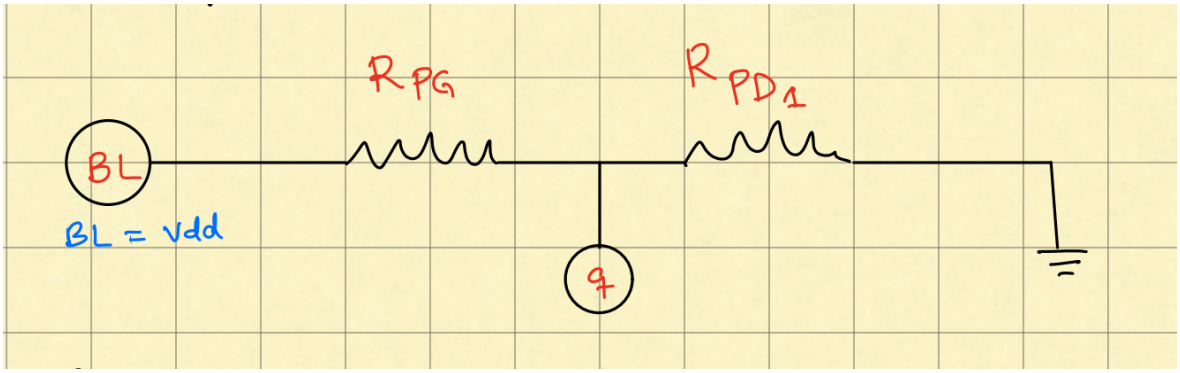


Figure 3: Original handwritten read-divider sketch.

For equal nominal channel length, a resistive first estimate gives

$$\frac{V_Q}{V_{DD}} = \frac{R_{\mathrm{PD}}}{R_{\mathrm{PG}} + R_{\mathrm{PD}}} \approx \frac{1}{1+\beta}, \qquad V_Q \approx \frac{V_{DD}}{1+\beta}. \tag{2}$$

At 0.8 V, this simple expression predicts 0.400, 0.320 and 0.267 V for $\beta = 1$, 1.5 and 2. These are *hand estimates*, much higher than the DC crossings in Table 1 because the devices are nonlinear and the ratio is only a proxy for their actual strengths. Their useful prediction is the downward trend with increasing β.

For a more physical but still ideal long-channel model, set $E = V_{DD} - V_{Tn}$ and $k = \mu_n C_{ox}(W/L)$. The pass-gate is saturated with $V_{GS} = V_{DS} = V_{DD} - V_Q$; the pull-down is in triode if $V_Q < E$. Neglecting leakage and body effect, Kirchhoff's law at Q gives

$$\underbrace{\tfrac{1}{2}k_{\mathrm{PG}}(E - V_Q)^2}_{I_{\mathrm{PG}}} = \underbrace{k_{\mathrm{PD}}(EV_Q - \tfrac{1}{2}V_Q^2)}_{I_{\mathrm{PD}}}. \tag{3}$$

Solving the physical root using $k_{\mathrm{PD}}/k_{\mathrm{PG}} \approx \beta$ yields

$$V_Q = E\left(1 - \sqrt{\frac{\beta}{1+\beta}}\right). \tag{4}$$

The positive quadratic root violates the assumed operating region. The original handwritten steps in Figure 4 show the pass-gate current, current-ratio equation and closed form; the equations above state them legibly. Neither model substitutes for the GPDK simulation.

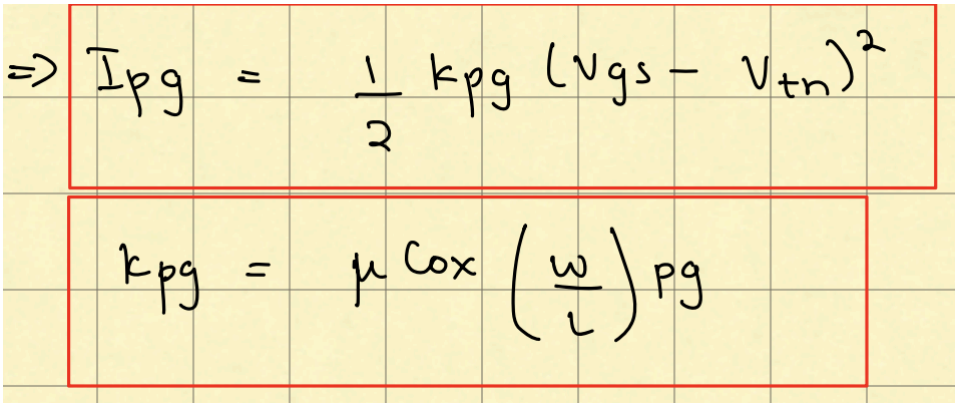


Figure 4: Original handwritten pass-gate current equation.

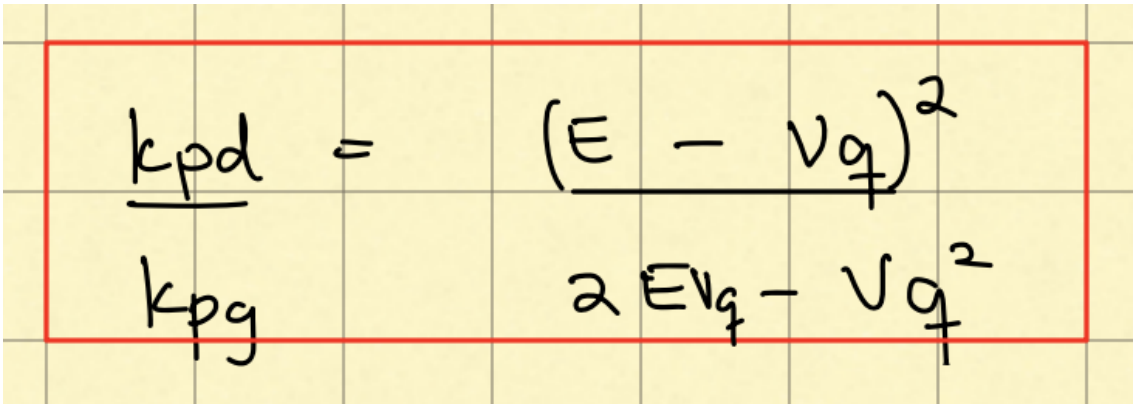


Figure 5: Original handwritten PD/PG current balance.

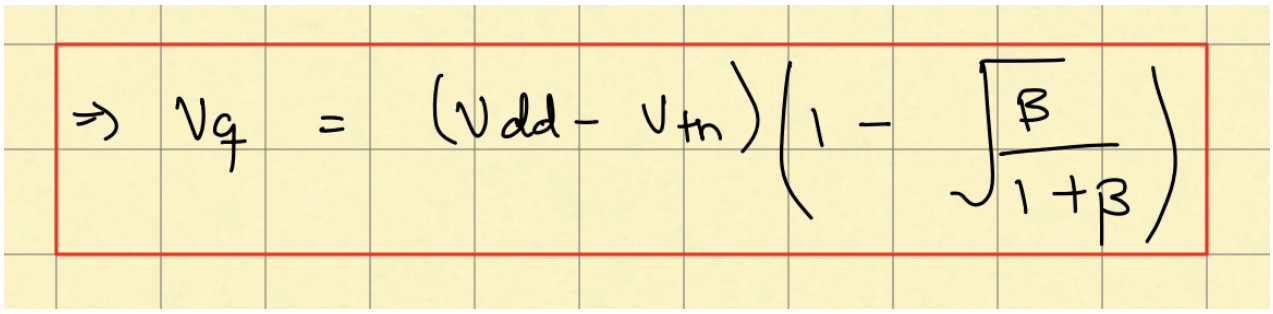


Figure 6: Original handwritten read-disturb voltage root.

The disturbed node drives INV2. The read criterion is $V_Q < V_M$, where V_M is INV2's switching voltage. In a square-law inverter approximation,

$$V_M = \frac{V_{Tn} + B(V_{DD} - |V_{Tp}|)}{1 + B}, \quad B = \sqrt{\frac{k_{\mathrm{PU}}}{k_{\mathrm{PD}}}} \approx \sqrt{\frac{\gamma\mu_p}{\beta\mu_n}}. \tag{5}$$

V_M is not independently measured here. Increasing β lowers the disturb voltage, but it also changes V_M, area and later bitline loading; measured read SNM is therefore the deciding schematic quantity.

3.2 Write contention and the pull-up ratio

For a write of zero into a stored one, take $Q \approx V_{DD}$ initially, $\overline{Q} \approx 0$, BL= 0, WL= V_{DD}. The pass-gate discharges Q against an ON pull-up. The first-order divider reverses the read case:

$$\frac{V_Q}{V_{DD}} \approx \frac{R_{\mathrm{PG}}}{R_{\mathrm{PG}} + R_{\mathrm{PU}}} \approx \frac{\mu_p\gamma}{\mu_n + \mu_p\gamma}. \tag{6}$$

A smaller γ weakens the pull-up relative to the pass-gate and lowers the written node. The receiving inverter must switch before the cell can settle back, so the condition is $V_Q < V_M$. If $\rho = \mu_n/\mu_p$, the divider implies $\gamma < \rho V_M/(V_{DD} - V_M)$, conditional on the assumed resistor model. Without a measured V_M or process strengths this is a design inequality, not a numeric proof that $\gamma = 1$ writes.

The more detailed hand current balance treats the pass-gate as triode and the pull-up as saturated while $V_Q < |V_{Tp}|$. With $E = V_{DD} - V_{Tn}$ and $F = V_{DD} - |V_{Tp}|$,

$$I_{\mathrm{PG}} = k_{\mathrm{PG}}(EV_Q - V_Q^2/2), \tag{7}$$

$$I_{\mathrm{PU}} = \tfrac{1}{2}k_{\mathrm{PU}}F^2, \qquad \frac{k_{\mathrm{PU}}}{k_{\mathrm{PG}}} \approx \gamma\frac{\mu_p}{\mu_n}, \tag{8}$$

$$V_Q = E - \sqrt{E^2 - \gamma(\mu_p/\mu_n)F^2}. \tag{9}$$

The other quadratic root is inconsistent with the assumed pass-gate region. Equation (9) also ceases to apply if the pull-up enters triode; no extracted model parameters have been inserted to make a fabricated numeric write limit. A stronger pass-gate helps writing but reduces β and adds load; the $\gamma = 1$ candidate needs a write-margin sweep before acceptance.

4 Read-SNM bench and reduction method

4.1 Circuit setup

The available Cadence benches break the inverter feedback for DC transfer curves. The Q sweep records $\overline{Q}$; the complementary sweep records Q. Both use WL= V_{DD}, BL=BLB= V_{DD} and a 0–0.8 V sweep. Figure 7 identifies the actual circuit setup rather than replacing it with a generic butterfly drawing.

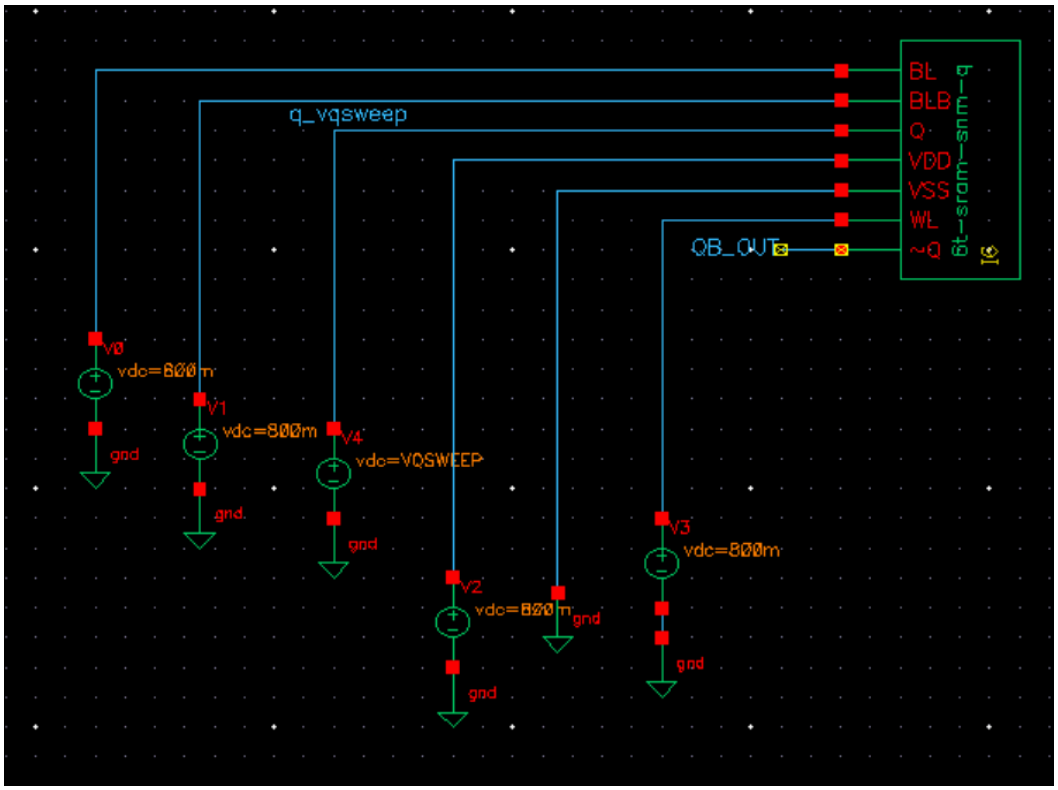


Figure 7: Original Virtuoso Q-side read-SNM sweep bench.

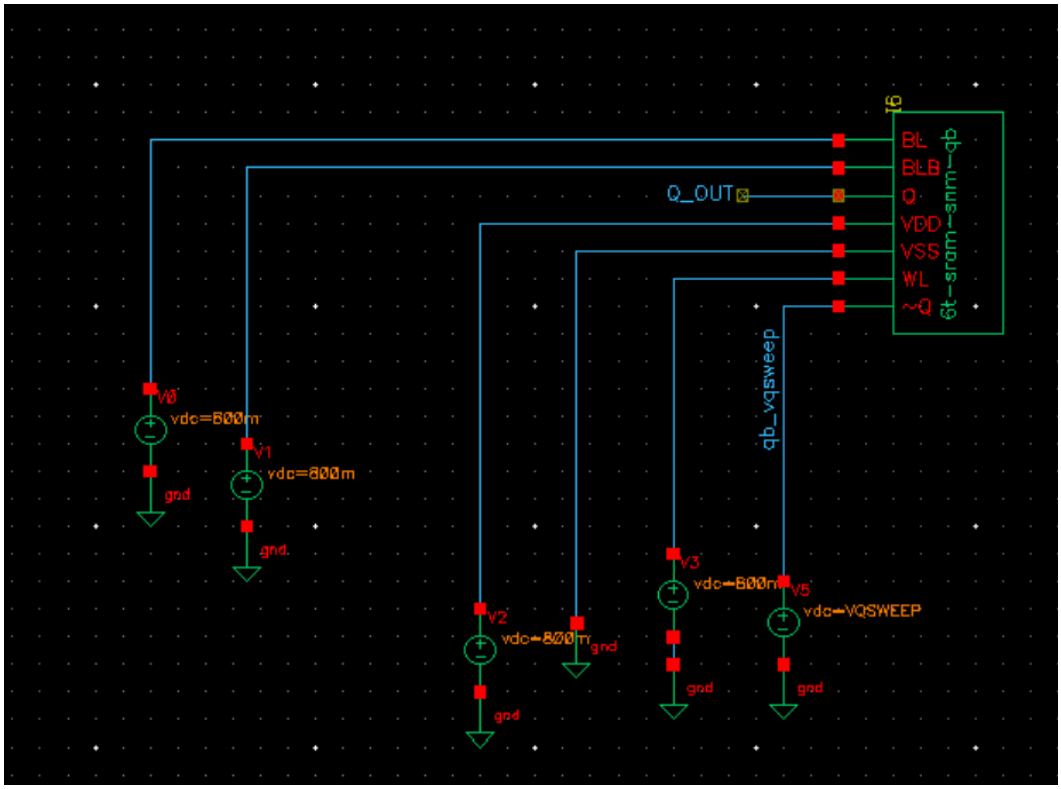


Figure 8: Original Virtuoso $\overline{Q}$-side read-SNM sweep bench.

4.2 From ViVA export to a margin

Each run has a four-column `raw.csv` ViVA export. `scripts/check_snm_sweep.py` detects which trace is normal and which is reflected, writes normalized `q.csv` and `qb.csv`, mirrors one VTC about $y = x$, locates the three intersections, and fits the largest axis-aligned square within each of the two lobes. The reported margin is

$$\mathrm{SNM}_{\mathrm{read}} = \min(\mathrm{SNM}_L, \mathrm{SNM}_R). \tag{10}$$

A 16,001-point voltage grid is used for the final result. Repeating the fit at 4,001 points changes the side length by at most 0.05 mV. The square coordinates and crossings are retained in each `snm.json`. The low-voltage crossing is reported as the DC "0"-node read-disturb operating point. This is a *schematic* analysis, even though Part 1 ultimately asks for the same measurements after extraction.

5 Measured schematic read stability

Table 2: Three-point read sweep at 0.8 V, from the raw Spectre exports. Extra digits preserve calculation traceability.

β	PD fins	SNM_L (mV)	SNM_R (mV)	$\mathrm{SNM}_{\mathrm{read}}$ (mV)	low crossing (mV)
1.0	2	126.6	126.6	126.6	134.3
1.5	3	145.8	145.8	145.8	96.6
2.0	4	156.3	156.3	156.3	77.1

For submission at the handout's requested two significant figures, the three read-SNM results are 0.13, 0.15 and 0.16 V; the corresponding disturbed zero-node crossings are 0.13, 0.097 and 0.077 V. The additional digits in Table 2 are retained to distinguish the underlying calculations. The selected candidate gains 29.7 mV (23%) over $\beta = 1$ at this schematic operating point. The low crossing drops by 57.2 mV. That direction agrees with Equations (2) and (4). No layout-area penalty or write-margin trade-off has been measured yet.

5.1 Original Cadence butterflies with fitted squares

Figures 9–11 are the *original ViVA screenshots*; the cyan outlines are report overlays placed at the coordinates saved by the maximum-square script. The voltage axes in ViVA use different horizontal and vertical pixel scales, so a square in volts appears rectangular on screen. The outlines are analytical annotations, not marks produced by Cadence. They make the two fitted lobes visible on the actual captured curves.

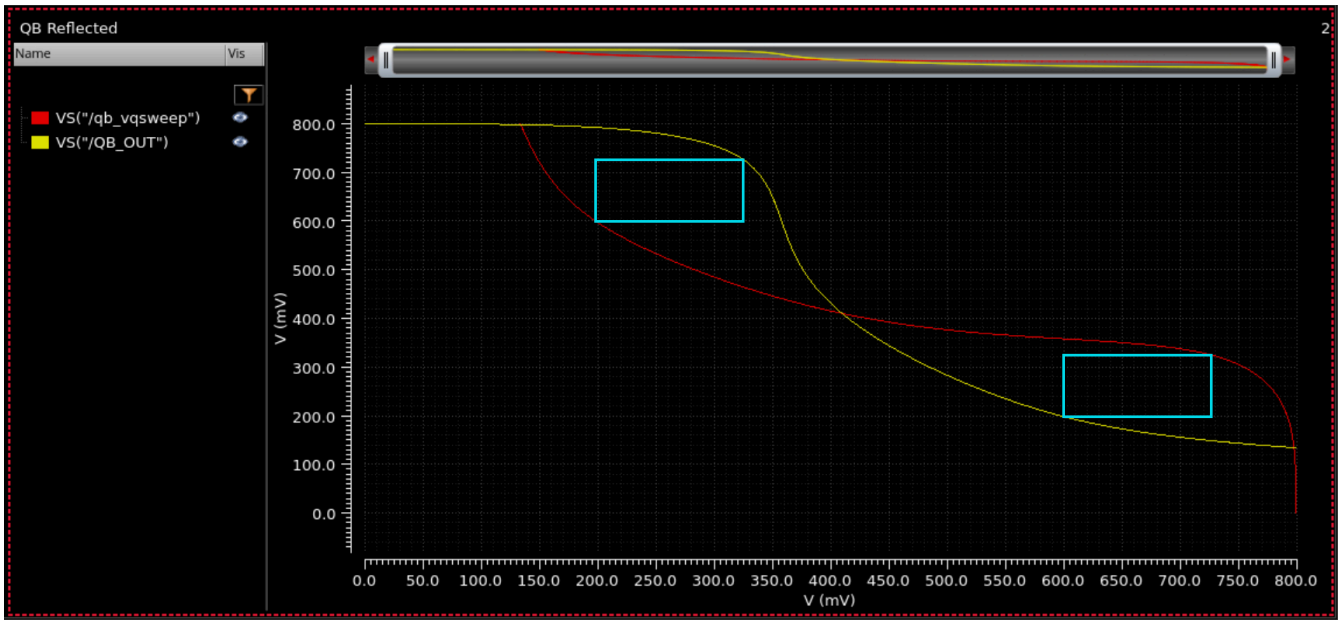


Figure 9: Original $\beta = 1$ ViVA butterfly with both computed largest-square locations overlaid; SNM = 126.6 mV.

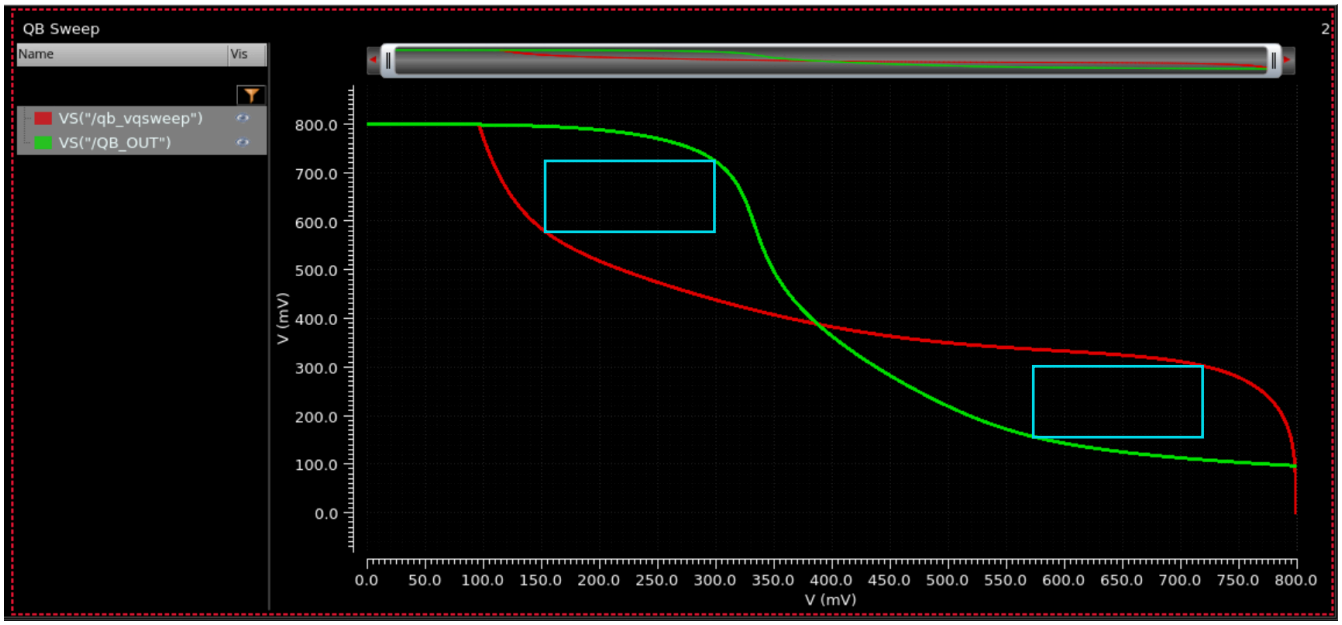


Figure 10: Original $\beta = 1.5$ ViVA butterfly with fitted-square overlays; SNM = 145.8 mV.

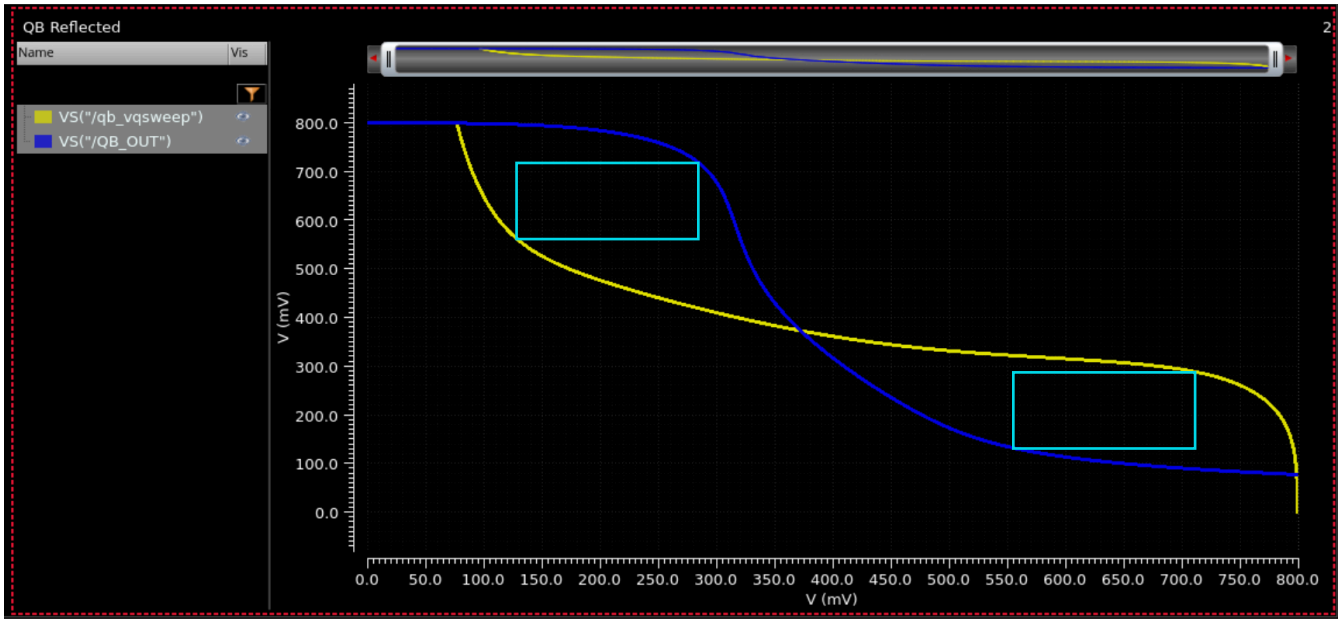


Figure 11: Original $\beta = 2$ ViVA butterfly with fitted-square overlays; SNM = 156.3 mV. This is the selected schematic candidate.

6 Layout and extraction: work state

The available `media/part1/6t_sram_layout_unlaid.png` shows separated device structures and is a planning artifact. It is useful as evidence that placement was explored, but it does not show the continuous, tileable finished cell demanded by the handout. It provides no reliable boundary dimensions, DRC or LVS result. Figure 12 is therefore labeled explicitly rather than reported as a completed layout.

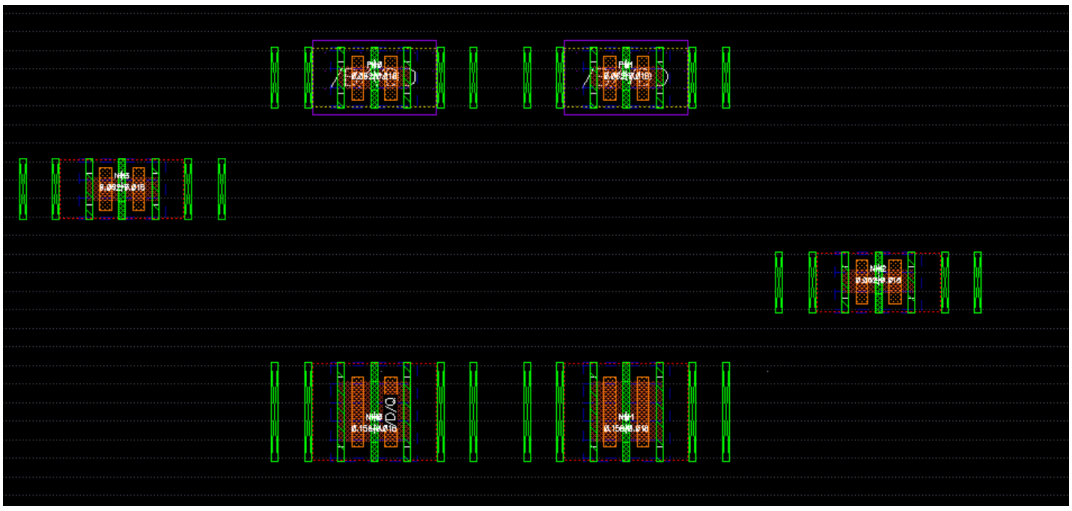


Figure 12: Unlaid placement draft. It is *not* a DRC/LVS-clean bitcell layout and cannot support area, parasitic or post-layout claims.

The final layout needs a horizontal wordline and vertical BL/BLB pair, with abutting cells able to share the intended bitline contacts and supply rails. After routing and signoff, the cell area must be measured as $A_{\mu m^2}$ and converted using the handout's $F = 10$ nm:

$$A_{F^2} = \frac{A_{\mu m^2}}{(0.01\,\mu\text{m})^2} = 10^4 A_{\mu m^2}. \tag{11}$$

No numeric area is printed because no finished boundary measurement exists. Likewise, the extracted per-cell BL-node and WL-node capacitances must be read from the completed parasitic view; the schematic sweep cannot supply them. A simple width-times-length transistor sum would not be a substitute for the required cell layout area.

7 Remaining Part 1 characterization

Table 3 answers the handout's remaining items with an explicit evidence state and the bench or artifact needed. These are action definitions, not claimed test outcomes. For any asymmetric butterfly, use the smaller of the two fitted square sides.

Table 3: Required post-layout and functional results. All numeric cells remain unreported until the extracted benches run.

Item	Required setup / report artifact	Current state
Hold SNM	WL= 0; two broken-feedback inverter VTCs; butterfly and $\min(\mathrm{SNM}_L, \mathrm{SNM}_R)$.	**Not yet evidenced**
Read SNM	WL= V_{DD}, BL=BLB= V_{DD}; extracted butterfly and DC zero-node voltage.	Schematic only
Write margin	Store $Q = 1$, WL= V_{DD}; hold the opposite bitline at V_{DD} and sweep the write-zero bitline downward. Record flip-point $V_{\mathrm{trip}} = \mathrm{WM}_{\mathrm{BL}}$.	**Not yet evidenced**
Cell current	Read a stored zero with bitlines at V_{DD}; report current drawn from its bitline.	**Not yet evidenced**
Supply dependence	Repeat read SNM and write margin at 0.8, 0.7, 0.5 V; state whether the cell still reads and writes at 0.5 V.	0.8 V schematic read only
Functional check	Transient WL/precharge pulses and 15 extra per-cell bitline loads; write 0/read, then write 1/read. Show WL, BL, BLB, Q, $\overline{Q}$.	**Not yet evidenced**

The post-layout supply table cannot honestly be filled from the current files. In particular, the 0.8 V read number above must not be copied into its post-layout column. The bitline load for the transient column surrogate also depends on the missing extracted per-cell capacitance. No waveforms in the repository establish a non-disturbing read or completed write under that load.

8 Discussion and next decisions

The schematic sweep shows the expected read trade-off: strengthening PD relative to PG lowers the read-disturbed zero node and enlarges the butterfly margin. The candidate $\beta = 2$ is supported *only for schematic read stability at 0.8 V*. Its larger PD may cost cell area and change internal and bitline parasitics, while $\gamma = 1$ still needs a quantitative writability test. Part 2 should inherit a cell only after layout, DRC/LVS, PEX and the complete Part 1 stability/functional benches are recorded. If extracted read SNM degrades or write margin is too small, revise PD/PG/PU sizing and repeat both schematic and extracted checks. Per-cell C_{BL} and C_{WL} then become measured inputs to the 16-by-4 array's line-load estimate.

Traceability and team contributions

The handout used for this report is `docs/ese5700-project1-current.pdf` (Part 1 on pp. 3–4). Reproducible results are in `results/part1/read-snm/beta1/`, `beta1p5/`, and `beta2/`; each contains `raw.csv`, normalized curves and `snm.json`. The overlay coordinates are the two `lobes` entries in those JSON files. Recompute with the command below. The original handwritten and Cadence images are retained in `media/part1/`.

```
python3 scripts/check_snm_sweep.py -all
```

The selected design was also captured in separate Q- and $\overline{Q}$-side sweep schematics (Figure 13). These establish the nominally symmetric device assignment used by the two transfer-curve benches; they do not provide layout evidence.

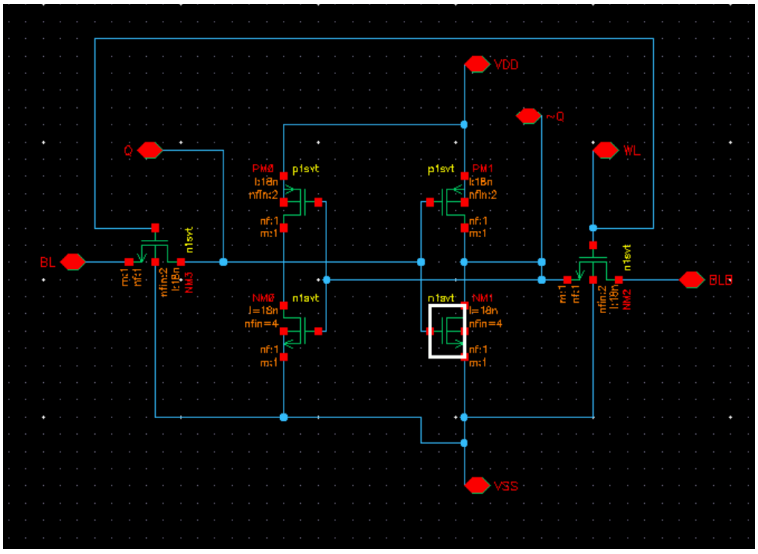


Figure 13: Original Cadence Q-side cell for $\beta = 2$.

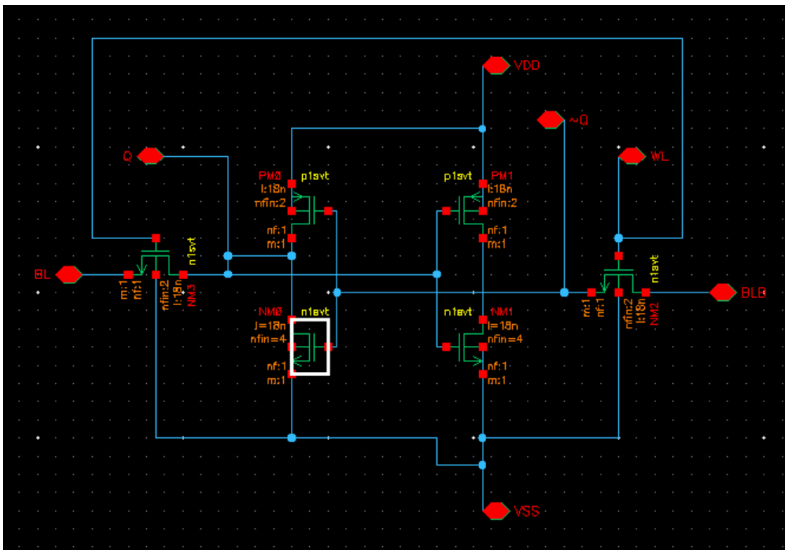


Figure 14: Original Cadence $\overline{Q}$-side cell for $\beta = 2$. The other PD variants remain in `media/part1/`.

Both authors are responsible for understanding the complete design and report. The repository does not state which author performed each task, so a contribution breakdown cannot be assigned without their input; this is an open handout requirement before team submission.